

San Francisco, CA

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Frederick Robinson

Employment

- 2024–2026 **Principal Applied Scientist (later Member of Technical Staff)**, *Amazon SF AI Lab*
- Founding member of the reinforcement learning group, setting up RL systems and advancing CoT/RLVR training.
 - Built [Turnstile](#) ([Amazon Science](#)), an open-source proxy capturing token-level rollout data for agentic RL training.
 - Drove data research for RLVR post-training: dataset ablations, offline difficulty calibration (16 samples against SOTA models), and targeted data acquisition against clustered eval failures.
- 2024 **Member of Technical Staff**, *Adept*
- Improved reliability of computer-use agents by developing evaluation frameworks, data collection pipelines, and SFT experiments.
 - Contributed foundational work that supported the team's transition into Amazon.
- 2022–2023 **Staff Machine Learning Engineer**, *Stripe, Inc. – Supportability Intelligence*
- Led an LLM-based enforcement initiative and a cross-functional team of ML and backend engineers, delivering > 20% improvement in Terms of Service violation detection.
 - Developed a transformer-based merchant website text classification model and a new service centralizing merchant business model data.
- 2017–2022 **Senior Applied Scientist**, *Twitch (Amazon AWS Interactive Video Service)*
- Tech lead for Video Intelligence. Built production ML at scale for data center selection (> 1% global buffering reduction) and real-time viewbot detection (> 75%).
- 2014–2017 **Software Engineer**, *Google, Inc. – Display Ads, Remarketing Quality*
- Built ML models and real-time bidding systems that optimized advertiser ROI and unlocked tens of millions in incremental annual revenue.

Skills

ML Large Language Models, Reinforcement Learning (RLHF, CoT, RLVR), Supervised Fine-Tuning, Evaluation Frameworks, Distributed Training Systems, Data Collection Pipelines

Tools PyTorch, slime, SGLang, Python, Golang

Education

2014 **Master of Arts in Mathematics**, *University of California Los Angeles*

2012 **Bachelor of Arts in Mathematics with Honors**, *Northwestern University*